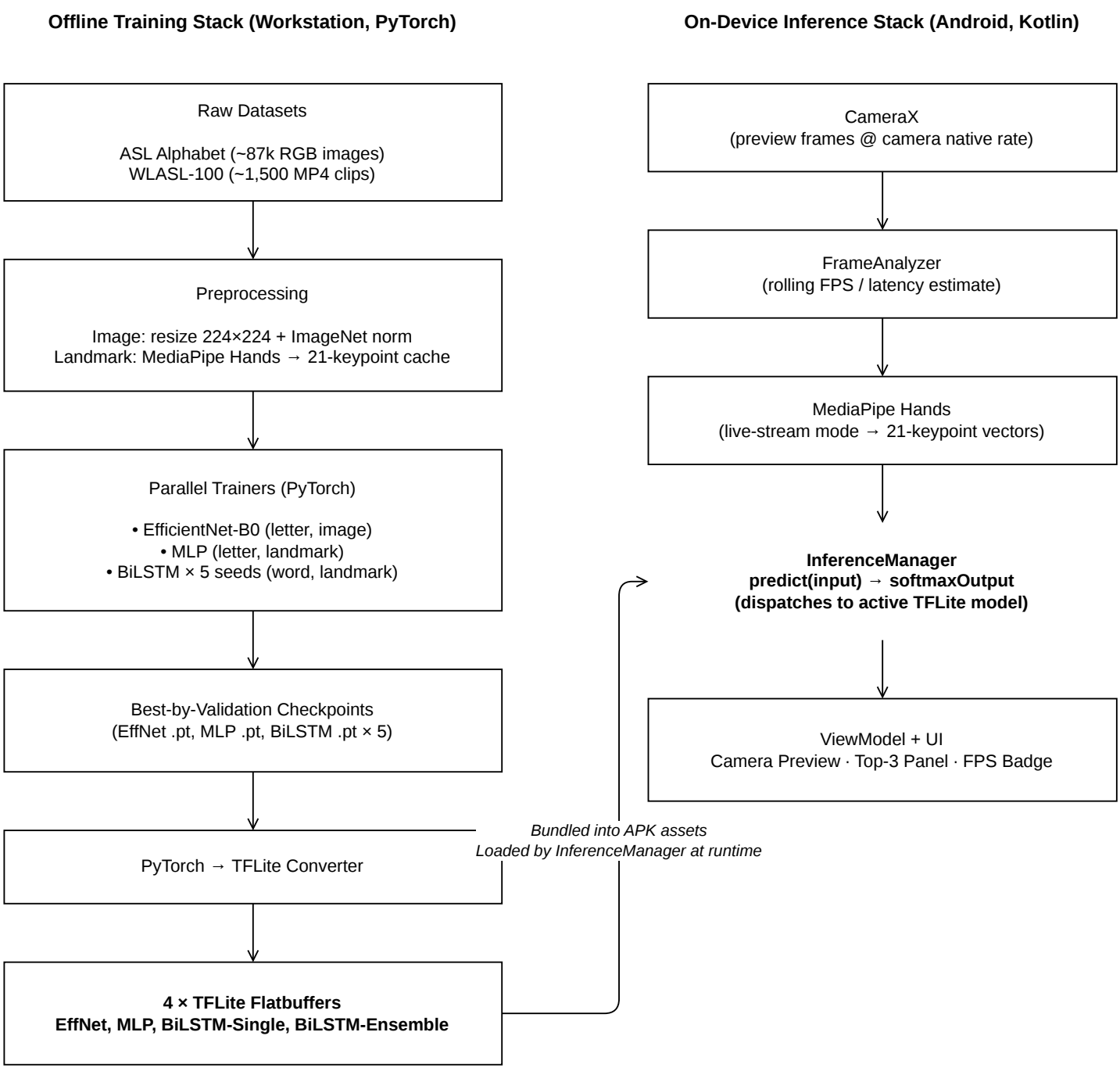
Offline Training Stack (Workstation, PyTorch)
On-Device Inference Stack (Android, Kotlin)
Raw Datasets
ASL Alphabet (~87k RGB images)
WLASL-100 (~1,500 MP4 clips)
Preprocessing
Image: resize 224×224 + ImageNet norm
Landmark: MediaPipe Hands → 21-keypoint cache
Parallel Trainers (PyTorch)
• EfficientNet-B0 (letter, image)
• MLP (letter, landmark)
• BiLSTM × 5 seeds (word, landmark)
Best-by-Validation Checkpoints
(EffNet .pt, MLP .pt, BiLSTM .pt × 5)
PyTorch → TFLite Converter
4 × TFLite Flatbuffers
EffNet, MLP, BiLSTM-Single, BiLSTM-Ensemble
Bundled into APK assets
Loaded by InferenceManager at runtime
CameraX
(preview frames @ camera native rate)
FrameAnalyzer
(rolling FPS / latency estimate)
MediaPipe Hands
(live-stream mode → 21-keypoint vectors)
InferenceManager
predict(input) → softmaxOutput
(dispatches to active TFLite model)
ViewModel + UI
Camera Preview · Top-3 Panel · FPS Badge